Assignment – 08:

Problem Statement:

Write a program in C to accept marks of five courses of a student and compute the result. A student is considered PASS if he/she scores 40 marks or more in each course. If the student passes, calculate the aggregate percentage and assign the grade as follows:

- Aggregate ≥ 75% : Distinction
- Aggregate ≥ 60% and < 75% : First Division
- Aggregate ≥ 50% and < 60% : Second Division
- Aggregate ≥ 40% and < 50% : Third Division

Code:

```
#include <stdio.h>
int main()
{
    int m1,m2,m3,m4,m5;
    float avg;
    printf("Enter marks of five subjects: ");
    scanf("%d%d%d%d%d",&m1,&m2,&m3,&m4,&m5);
    printf("Total Marks: %d\n",m1+m2+m3+m4+m5);
    avg=(m1+m2+m3+m4+m5)/5;
    printf("Percentage: %f\n",avg);
    if (m1>=40&&m2>=40&&m3>=40&&m4>=40&&m5>=40)
    {
        printf("Result: Pass\n");
        if (avg>=75)
        {
            printf("Grade: Distinction");
        }
        else if ((avg>=60)&&(avg<75))
        {
```

```
                printf("Grade: First Division");
            }
            else if ((avg>=50)&&(avg<60))
            {
                printf("Grade: Second Division");
            }
            else if ((avg>=40)&&(avg<50))
            {
                printf("Grade: Third Division");
            }
        }
        else
        {
            printf("Result: Fail");
        }
}
```

Input & Output:

```
Enter marks of five subjects: 98 99 95 96 97
Total Marks: 485
Percentage: 97.000000
Result: Pass
Grade: Distinction
```

```
Enter marks of five subjects: 67 56 87 92 65
Total Marks: 367
Percentage: 73.000000
Result: Pass
Grade: First Division
```

```
Enter marks of five subjects: 54 53 47 52 51
Total Marks: 257
Percentage: 51.000000
Result: Pass
Grade: Second Division
```

```
Enter marks of five subjects: 43 45 46 47 48
Total Marks: 229
Percentage: 45.000000
Result: Pass
Grade: Third Division
```

```
Enter marks of five subjects: 56 45 42 39 45
Total Marks: 227
Percentage: 45.000000
Result: Fail
```